

Structural Mode N in CPTG

Galaxy Structural Classification from the Solved CPTG Field

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Abstract

Conventional galaxy classification is appearance-first: observations of the luminous galaxy are interpreted and an external catalog assigns a structural label such as a Hubble type or morphology family. Curvature Polarization Transport Gravity (CPTG) reverses that direction. The galaxy is first solved from its own resolved kinematic and baryonic source measurements. Structural Mode N is then derived from the radial organization of that solved field, and the Curvature Structure Mode Index (CSMI) maps the resulting mode to a structural class. No Hubble type, morphology family, or visual judgment of what the galaxy “looks like” enters the calculation of N . For CPTG structural classification, the classification authority therefore moves from an external morphology catalog to the solved galaxy itself.

This solution-first classification is tested on the primary 153-galaxy SPARC sample. Empirical N strongly orders the non-input SPARC Hubble-type sequence, with Spearman $\rho_S = -0.60817$ and $p = 7.58 \times 10^{-17}$. The ordering remains significant after explicit controls for radial point count and distance and remains nearly unchanged in progressively better-resolved subsets. N also follows non-input surface-brightness, size, luminosity, gas-content, and flat-velocity structure, while the inherited CSMI bands recover the same broad population progression from irregular and late-type systems toward earlier structural populations.

The principal result is not a morphology fit. It is that CPTG can generate a reusable structural catalog from the solved galaxies without importing morphology as an input. Observational measurements remain essential: photometry, gas measurements, and kinematics supply the quantitative source information used to solve the galaxy. What is removed from the classification step is the externally assigned appearance label. CSMI therefore makes structural class an output of the solved CPTG field rather than a prior description supplied to it. The present evidence is a primary-SPARC validation, with prospective cross-survey replication reserved for a future comparably resolved independent sample.

1 Introduction

Conventional galaxy morphology is appearance-first. A galaxy is imaged, its luminous structure is interpreted, and an external catalog assigns a label such as irregular, spiral, lenticular, or elliptical. Historically those classifications were made by individual astronomers or small expert teams; modern catalogs also use crowdsourced visual classifications and automated pipelines. These approaches can be highly useful and statistically rigorous, but the structural label remains an interpretation of observed appearance rather than a quantity generated by solving the galaxy’s gravitational field [2, 3].

Appearance-based classification is also observationally contingent. The structures visible to a classifier depend on wavelength, image depth, angular resolution, inclination, obscuration, and the classification procedure itself. Visual catalogs therefore carry classifier uncertainty and bias, and automated classifiers can inherit the assumptions or labels of the training set on which they are based [2, 3]. This is not an argument against observational catalogs; they are indispensable records of measured light and other galaxy properties. It is a statement about where the authority for the structural label originates: conventionally, it originates outside the gravitational solution.

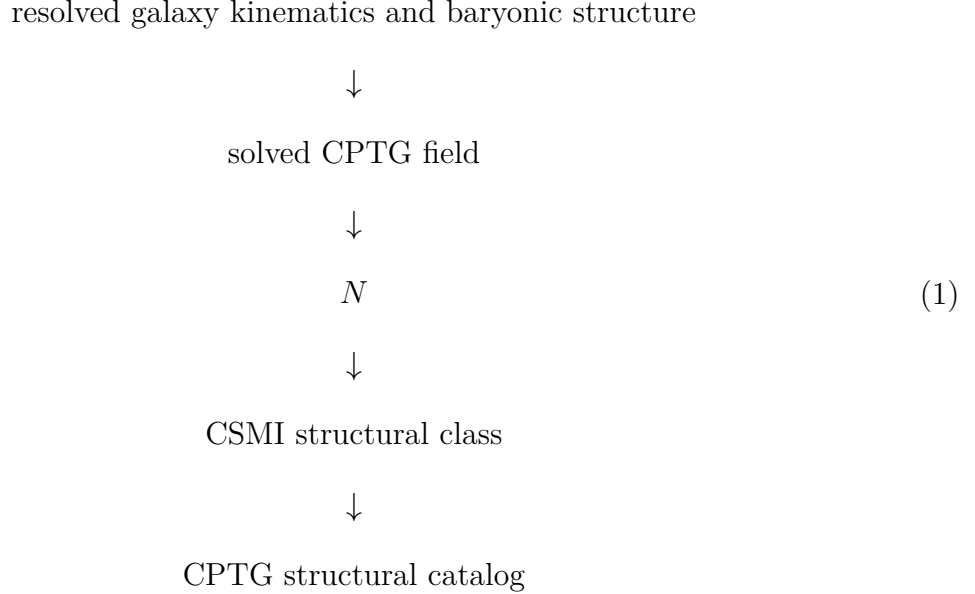
Structural Mode N changes that direction inside Curvature Polarization Transport Gravity (CPTG) [10, 11]. A CPTG galaxy is first solved from its own resolved rotation curve and baryonic source structure. Only after that field solution is retained is Structural Mode evaluated. N is a dimensionless readout of where radial change in the solved curvature-polarization/transport structure is concentrated. Low modes correspond to organization weighted farther toward the outer part of the registered radial interval; progressively larger modes correspond to organization weighted farther inward. Thus N is neither an observed morphology label nor another rotation-curve fit parameter. It is information extracted from the solved galaxy itself.

The distinction is strict. No variable in the Structural Mode calculation asks whether the galaxy should be visually classified as barred, spiral, irregular, bulge-dominated, or any other morphology. Quantitative disk and bulge source contributions can enter the baryonic source model through measured inputs, but a morphology label for those components does not. Once the resolved kinematic and baryonic inputs are fixed, what an observer calls the galaxy has no mathematical role in the resulting N . Two researchers using the same resolved inputs and the same CPTG implementation should recover the same Structural Mode regardless of the morphology label either researcher would have assigned by eye.

The Curvature Structure Mode Index (CSMI) turns that solved coordinate into a structural classification. CSMI assigns descriptive structural labels to the current catalog intervals of N , allowing a population of solved galaxies to be organized into a CPTG structural catalog without supplying conventional morphology to the field calculation. This is the central predictive role of Structural Mode: a galaxy with no morphology label supplied to CPTG can still classify itself within the same structural coordinate system as every other solved galaxy. Researchers can therefore compare like-mode populations, identify structural transitions and outliers, and construct mode-selected samples from the solved field rather than from an externally imposed appearance class.

For CPTG, the role of the catalog is therefore reversed. Conventional morphology is an input description available before the gravitational analysis; CSMI is a derived output available only after the galaxy has been solved. Observational catalogs remain necessary to supply the measured galaxy data, but they no longer have to tell CPTG what structural class the solved object belongs to.

The inference direction tested in this paper is therefore



No catalog Hubble type, morphology family, luminosity class, surface-brightness class, or other CSMI validation label is supplied along this chain. The empirical question is whether the classification produced at the end of the chain nevertheless recovers recognizable galaxy organization. In this paper, conventional morphology is therefore not the classification authority; it is a non-input validation target against which the theory-generated catalog can be tested.

SPARC provides a direct internal test because its resolved rotation curves are accompanied by catalog quantities such as Hubble type, luminosity, effective radius, surface brightness, disk scale length, neutral-hydrogen mass, and flat rotation velocity [1]. The calculations reported here are executed with the CPTG SPARC Browser Workbench v1.12.0, introduced in Section 2. The workbench uses the rotation-curve and baryonic-source files to solve each galaxy; the catalog structural quantities are retained downstream as non-input validation coordinates. The primary test is therefore not whether N can be fit to morphology, but whether a structural coordinate calculated without morphology reproduces an independently assigned population ordering afterward.

The distinction also defines the evidentiary boundary. The rotation curves and validation metadata belong to the same SPARC observational program, so this paper establishes a primary-SPARC structural validation rather than a cross-survey replication. Radial sampling is treated explicitly because the registered sampled-data estimator is sensitive to the resolved radial grid. The unchanged Structural Mode definition is tested under point-count and distance controls, measurement perturbations, radial resampling, and progressively better-resolved subsamples. No conventional analogue for N is assumed in

the analysis; its scientific value is judged operationally by whether the galaxy’s own solved field generates a stable structural catalog that agrees with non-input observations.

2 CPTG SPARC Browser Workbench

The CPTG SPARC Browser Workbench [11] is the registered local analysis environment used in this paper to apply the same galaxy calculation consistently across the SPARC database. It loads the resolved rotation-curve and baryonic-source files, executes the CPTG galaxy solution, evaluates the sampled Structural Mode estimator and CSMI output, and keeps catalog morphology and the other validation coordinates outside the field-solution inputs. That separation implements the inference chain in Eq. (1): conventional morphology can be displayed and compared after the solve, but it does not determine the calculated N .

The workbench is also a reader-facing research interface rather than a manuscript-only execution script. It supports direct single-galaxy and batch analysis, displays solved CPTG and comparison rotation curves, exposes the relevant galaxy meta-data, reports Structural Mode and CSMI results, and can save selected numerical and graphical outputs when a persistent record is required. Its purpose is reproducibility and exploration: a reader can move from the population results in this paper to an individual galaxy without reconstructing the analysis pipeline by hand.

The software does not add another theoretical layer. The field equation, the galaxy structural solution, the Structural Mode definition, and the CSMI readout are defined independently of the interface and are stated explicitly in the sections that follow.

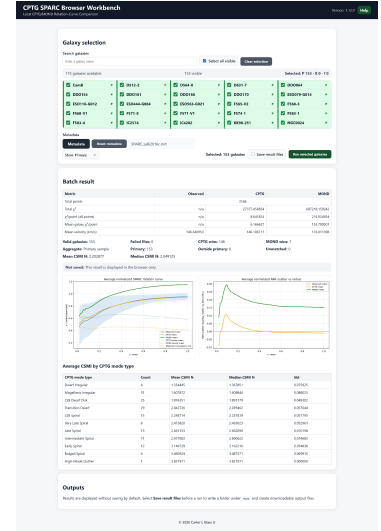


Figure 1: CPTG SPARC Browser Workbench in batch-analysis mode.

3 The CPTG Structural Acceleration Scale $a_\star$

Before the galaxy field equation is introduced, the role of $a_\star$ must be separated from the pointwise accelerations that appear in a rotation curve. In CPTG, $a_\star$ is the structural acceleration scale of the gravitating object. It is an object-level property of the solved gravitational state, not the centripetal acceleration V^2/r at one radius, not the baryonic acceleration $g_{\text{bar}}(r)$, and not the final solved field $g(r)$ itself. Across CPTG the same quantity can be inferred from resolved dynamics, derived inside another structural reduction, selected by a boundary-value solution, or remain observationally unresolved; those are different ways of determining one structural quantity rather than different definitions of it [10].

The assignment is deliberately non-universal. In standard GR/ Λ CDM galaxy modelling there is no corresponding object-specific structural acceleration coordinate in the gravi-

tational law; the additional gravitational support is conventionally represented through a non-baryonic halo component. MOND does introduce an acceleration scale, a_0 , but assigns one universal constant to all galaxies [4]. CPTG instead allows each gravitating object to carry its own a_* . The analogy to MOND is therefore only that both quantities have the dimensions and dynamical role of an acceleration scale; a_* is not MOND’s a_0 and is not required to take the same value from galaxy to galaxy.

Nor is a_* a cosmological expansion parameter. In a quasi-static galaxy calculation such as the present one, it is the structural value associated with the galaxy at the solved observational state. In an explicitly dynamical problem an $a_*(t)$ may evolve with the structure of that object, but CPTG does not assume that evolution follows the Hubble rate, a universal cosmic clock, or the native cosmological π -branch acceleration anchor. The galaxy-scale a_* is therefore local to the object’s structural solution even though the same CPTG framework is also applied on cosmological scales [10].

The full CPTG structural domain is

$$a_* \in [0, \infty). \quad (2)$$

The present SPARC solver occupies the positive galaxy branch, $a_* > 0$. The endpoint $a_* = 0$ is the continuous zero-structural-response limit of the same CPTG quantity; formulas that divide by a_* are therefore used only on the positive branch.

For the positive galaxy branch used by the present solver, the structural scale determines the associated curvature-transport radius,

$$R_c = 48\pi \frac{c^2}{a_*}. \quad (3)$$

Thus changing a_* changes the structural radius together with the nonlinear response of the galaxy; it is not a free amplitude applied after the field has been solved. The implementation therefore tests trial values of a_* , derives the corresponding R_c for each trial, re-solves the full radial field, and retains the value selected by the resolved kinematic likelihood. Structural Mode N is evaluated only after that solution has been fixed. This ordering is central to the paper: a_* determines the object’s acceleration-response scale, whereas N measures how the retained solved response is organized radially.

4 Structural Mode Definition

4.1 Galaxy field invariant

For the solved radial CPTG acceleration field $g(r)$, define the structural field invariant

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad (4)$$

with the current galaxy implementation using

$$\kappa_{\text{struct}} = 1 \text{ kpc.} \quad (5)$$

Both $\kappa_{\text{struct}} = 1 \text{ kpc}$ and the outer readout interval $R/5 < r \leq R$ below are fixed features of the registered galaxy Structural Mode readout. They are held constant across all galaxies and all ablation branches in this paper; neither is adjusted galaxy by galaxy or selected from catalog morphology during the validation.

The post-solution radial weight is

$$w(r) = \left| \frac{dC_g}{dr} \right|. \quad (6)$$

Let R denote the outer solved radius. The registered outer organization length is the weighted centroid

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad (7)$$

and Structural Mode is

$$N = \frac{R}{\lambda}. \quad (8)$$

When w is not identically zero on the registered readout interval,

$$1 \leq N \leq 5. \quad (9)$$

If $w \equiv 0$ throughout the readout interval, the denominator in Eq. (7) vanishes and N is undefined rather than assigned an arbitrary structural mode.

4.2 Exact bounds and conditional invariances

The bound in Eq. (9) is independent of the amplitude of the solved field. Structural Mode is invariant under affine rescaling $C_g \rightarrow AC_g + B$ with $A \neq 0$, because the common factor $|A|$ cancels from Eq. (7). It therefore measures radial organization rather than the normalization or additive offset of the solved invariant.

It is also invariant under uniform radial similarity. Let

$$\tilde{r} = sr, \quad \tilde{R} = sR, \quad \tilde{C}_g(\tilde{r}) = AC_g(\tilde{r}/s) + B, \quad (10)$$

with $s > 0$ and $A \neq 0$. Then

$$\tilde{w}(\tilde{r}) = \frac{|A|}{s} w(\tilde{r}/s), \quad \tilde{\lambda} = s\lambda, \quad \tilde{N} = N. \quad (11)$$

Thus, for profiles related by the stated similarity transformation, the continuous mode is unchanged by an overall field rescaling, additive offset, or uniform rescaling of the resolved radial domain. This is a conditional invariance of the Structural Mode functional: because the operative galaxy invariant contains the fixed physical scale $\kappa_{\text{struct}} = 1 \text{ kpc}$, an arbitrary physical resizing of a galaxy does not automatically imply the transformation in Eq. (10). These properties make N a dimensionless organization coordinate rather than a proxy for field amplitude or galaxy size alone.

4.3 Operative galaxy solution

The Structural Mode readout is downstream of the fixed CPTG galaxy solution rather than a separate phenomenological fit. In the implementation tested here, the baryonic source field is constructed from the signed velocity contributions through

$$g_{\text{bar}}(r) = \frac{1}{r} \max \left\{ 0, \frac{27}{50} Q(V_{\text{gas}}) + \frac{23}{50} Q(V_{\text{disk}}) + \frac{16}{25} Q(V_{\text{bulge}}) \right\}, \quad Q(V) = V|V|. \quad (12)$$

For every trial a_* , R_c is derived and the nonlinear acceleration is solved from

$$g = g_{\text{bar}} + \frac{\sqrt{a_* g_{\text{bar}}} + g_{\text{geom}}}{[1 + (g/a_*)^{123/50}]^{11/50}}, \quad (13)$$

with

$$g_{\text{geom}} = \frac{1}{2} a_* \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}. \quad (14)$$

Here $R_c(a_*)$ is the curvature-transport radius already introduced in Eq. (3); it is derived from the trial a_* rather than fitted independently.

The retained value of a_* minimizes

$$\chi^2(a_*) = \sum_i \left[\frac{V_{\text{obs},i} - \sqrt{g(r_i; a_*)} r_i}{\sigma_{V,i}} \right]^2. \quad (15)$$

The source weights, exponents, transport-remnant coefficient, and $R_c(a_*)$ relation are fixed across the sample. No CSMI quantity appears in Eqs. (12)–(15).

4.4 Relation of Structural Mode to the acceleration scale

Section 3 introduced a_* as the object-specific structural acceleration scale and established that it is fixed by the galaxy solution before Structural Mode is read out. The operative equations above make that ordering explicit: each trial value has its own derived R_c , each trial requires a new nonlinear field solution, and only the retained solution is passed to C_g , w , λ , and N . Structural Mode therefore does not re-fit or redefine a_* ; it measures the radial organization of the response selected by the preceding kinematic solve.

This separation supplies a sharp empirical discriminator. If N and $a_\star$ encode the same information, N should track $a_\star$ strongly and should add little structural information after $a_\star$ is known. Section 8 tests that possibility directly.

4.5 Curvature Structure Mode Index

The Curvature Structure Mode Index (CSMI) attaches descriptive interval labels to the continuous-valued solved mode. The labels are downstream catalog names; they do not enter Eq. (8), the field equation, the $a_\star$ optimization, or the primary-sample selection.

The interval map used here is inherited unchanged from the pre-existing CPTG SPARC Workbench v1.12.0 implementation and is held fixed for the validation reported in this manuscript. That does not make the present CSMI taxonomy final. The current intervals are the registered catalog used to test the already-defined mode populations in this sample, not a claim that all possible structural populations have already been resolved. The present paper does not establish a morphology-independent historical derivation of the numerical band boundaries. Accordingly, the empirical N – T relation is treated as the threshold-independent structural validation, while CSMI-band versus catalog-morphology statistics are reported as descriptive concordance of the inherited map rather than as an independent derivation of those thresholds.

Structural Mode N	Descriptive CSMI label
$N \leq 1.45$	Dwarf Irregular
$1.45 < N \leq 1.75$	Magellanic Irregular
$1.75 < N \leq 1.95$	LSB Dwarf Disk
$1.95 < N \leq 2.15$	Transition Dwarf
$2.15 < N \leq 2.35$	LSB Spiral
$2.35 < N \leq 2.55$	Very Late Spiral
$2.55 < N \leq 2.80$	Late Spiral
$2.80 < N \leq 3.05$	Intermediate Spiral
$3.05 < N \leq 3.30$	Early Spiral
$3.30 < N \leq 3.55$	Bulged Spiral
$3.55 < N \leq 3.72$	Lenticular/Early Disk
$N > 3.72$	High-Mode Outlier

Table 1: Current CSMI interval map used for the present validation. Structural Mode N is the continuous-valued structural coordinate; the descriptive label is assigned only afterward. The map is a present catalog of resolved mode populations rather than a permanently closed taxonomy.

The descriptive names intentionally use familiar structural language because they make the solved mode immediately interpretable and testable. They are not asserted to be one-to-one replacements for catalog Hubble type. The empirical N – T relation is the threshold-independent structural test; the inherited interval map is evaluated as descriptive concordance. No conventional morphology is supplied to the field calculation or to the numerical evaluation of N .

Operationally, CSMI is not an appearance-based lookup table. Its classification input is the solved Structural Mode N , not an image, Hubble type, or morphology family. The same solved value of N receives the same CSMI class regardless of how the object was visually described beforehand. Object-level CSMI assignment in the registered map therefore depends only on solved N ; the historical provenance of the present named interval boundaries is a separate taxonomy-level question. This is the mechanism by which CPTG can populate its own structural catalog from solved galaxies while reserving external morphology for validation rather than classification authority.

As larger and more diverse galaxy samples are analyzed, future CSMI catalogs will likely expand, subdivide, or refine the named structural types to represent newly resolved mode populations. Any additional categories would remain derived from the solved CPTG mode distribution rather than being imposed from conventional morphological classifications. The defining quantity therefore remains continuous Structural Mode N ; CSMI is the evolving catalog layer built from the populations that emerge in that solved coordinate. Prospective replication of the present CSMI map should apply the registered intervals in Table 1 unchanged. Any later subdivision, expansion, or renaming would constitute a new CSMI catalog derived from the enlarged solved-mode population and should be validated separately.

The direct research-facing use is therefore

$$\begin{array}{ccc}
 \text{solved CPTG galaxy} & & \\
 \downarrow & & \\
 N & & \\
 \downarrow & & (16) \\
 \text{CSMI descriptive band} & & \\
 \downarrow & & \\
 \text{cross-galaxy structural comparison} & &
 \end{array}$$

The label supplies an immediate common language for comparing galaxies with similar solved organization, while the empirical mode value retains the quantitative information needed to test gradients within or across those bands.

4.6 Exact numerical realization used in the validation

Equations (4)–(8) define the continuous Structural Mode. The v1.12.0 workbench evaluates the same readout on the discrete, generally nonuniform SPARC radial grid as follows. The radii are sorted, dg/dr is evaluated with the numerical gradient operator on that grid, and

$$C_i = g_i^2 + \kappa_{\text{struct}}^2 (D_r g)_i^2, \quad u_i = |(D_r C)_i|. \quad (17)$$

Only points satisfying $r_i > R/5$ are retained. Let n_{out} be the number of retained outer

points and let $\mathcal{S}_5$ denote the registered five-point moving-average filter. The weights used by the sampled estimator are

$$w_i = \begin{cases} (\mathcal{S}_5 u)_i, & n_{\text{out}} > 5, \\ u_i, & n_{\text{out}} \leq 5. \end{cases} \quad (18)$$

The implemented centroid and mode are then

$$\lambda_{\text{num}} = \frac{\sum_i r_i w_i}{\sum_i w_i}, \quad N_{\text{num}} = \frac{R}{\lambda_{\text{num}}}. \quad (19)$$

Equation (19) is the registered sampled-data estimator used by the workbench; it is not presented as a quadrature rule for Eq. (7), because the discrete sums do not introduce explicit radial-cell factors Δr_i . For nonnegative w_i it retains the centroid bound, $1 \leq N_{\text{num}} < 5$, whenever the denominator is nonzero, and a common affine rescaling of the sampled C_i leaves the weights proportional and therefore leaves N_{num} unchanged. It is not, however, invariant to changes in radial sampling density. This distinction is why radial-resampling and sampling-density controls in Section 6 are part of the validation: they assess the registered discrete readout without redefining the continuous functional in Eq. (8).

Hereafter, empirical N denotes the continuous-valued sampled estimator N_{num} unless the text explicitly refers to the continuous functional in Eq. (8). The phrase “continuous-valued” distinguishes the numerical mode value from the categorical CSMI label; it does not identify Eq. (19) with the continuum integral.

5 Data and Fixed Validation Protocol

5.1 Data-role firewall

The validation keeps the field-solution inputs, derived CPTG quantities, and catalog validation coordinates in distinct roles. Table 2 states that separation explicitly.

Field-solution inputs	Fitted or derived CPTG quantities	Non-input validation metadata
r	$a_\star$	Hubble type T
V_{obs}, σ_V	R_c	$L_{3.6}$
V_{gas}	solved $g(r)$	$R_{\text{eff}}, SB_{\text{eff}}$
V_{disk}	C_g, w, λ	$R_{\text{disk}}, SB_{\text{disk}}$
V_{bulge}	empirical N_{num} and CSMI label	$M_{\text{HI}}, R_{\text{HI}}, V_{\text{flat}}$
Not used as field input	Not CPTG-derived	D and radial point count n_{pt}

Table 2: Data-role firewall for the Structural Mode analysis. Catalog structural metadata and sampling descriptors in the right column do not enter the galaxy field solution, the numerical Structural Mode estimator, or the CSMI assignment. Inclination i and quality flag Q are used only for the primary-sample selection in Eq. (20); inclination uncertainty is used only in the optional inclination-systematic diagnostic.

The firewall is computational rather than a claim of observational independence. Catalog distance D is not passed as an explicit field-solution variable in this workbench implementation, but the published SPARC physical radii and photometrically derived mass-model quantities inherit the adopted observational distance calibration. Accordingly, D is treated here as non-input validation and sampling metadata at the workbench level while its upstream role in constructing the published SPARC physical data remains part of the dataset provenance [1].

5.2 Primary SPARC sample

The analysis begins from the 175 SPARC rotation-curve files used by the CPTG workbench and the official SPARC master metadata [1]. The primary science sample follows the workbench rule

$$Q \neq 3, \quad i \geq 30^\circ, \quad (20)$$

where Q is the SPARC quality flag and i is inclination. This leaves 153 galaxies containing 3168 usable rotation-curve points. The resulting primary-sample Structural Mode distribution has

$$\langle N \rangle = 2.20288, \quad \text{median}(N) = 2.04912, \quad 1.23536 \leq N \leq 3.82197. \quad (21)$$

The following metadata are retained only for downstream validation: Hubble type T , $3.6\,\mu\text{m}$ luminosity $L_{3.6}$, effective radius R_{eff} , effective surface brightness SB_{eff} , disk scale length R_{disk} , disk surface brightness SB_{disk} , neutral-hydrogen mass M_{HI} , HI radius R_{HI} , and measured V_{flat} . None is used to compute N . Inclination and quality flag are used only in Eq. (20); quoted inclination uncertainty enters only the optional inclination-systematic χ^2 diagnostic and does not enter CSMI. Catalog placeholder values $V_{\text{flat}} \leq 0$ are treated as unavailable rather than as physical zero velocities.

5.3 Fixed tests and statistical protocol

With the workbench equations, N definition, primary-sample rule, and CSMI thresholds held fixed, the validation addresses six distinct questions:

1. Does empirical N order non-input catalog morphology?
2. Does that ordering survive explicit controls for radial sampling density and distance?
3. Does N covary with non-input photometric and baryonic structural observables?
4. Does N add fold-held-out predictive information beyond declared metadata baselines, and how does that increment change when sampling descriptors are added?
5. Is the inferred N stable under measurement perturbation and radial resampling?
6. Is the structural ordering merely a surrogate for $a_\star$ or critically dependent on the galaxy-specific acceleration scale?

Spearman rank statistics are used for monotonic relations. For the principal N – T statistic, the 95% interval is generated from 5000 galaxy-level bootstrap resamples. Partial Spearman tests are obtained by rank-transforming the variables, regressing each target rank on the declared control ranks, and correlating the residuals. Multiple validation-metadata tests are corrected with the Benjamini–Hochberg false-discovery procedure.

The predictive-extension test treats the catalog T coordinate as an ordinal numerical response and uses a fixed ten-fold split with shuffled folds generated from random seed 20260829 and ridge coefficient $\alpha = 1$. Positive metadata and sampling predictors are transformed to base-10 logarithms and standardized inside each training fold. A declared metadata baseline is compared with the same model plus empirical N .

Because N is correlated with several baseline predictors, the primary conditional sensitivity uses a residual-permutation procedure rather than only an unconditional shuffle. Under the same fixed folds, the baseline predictors first generate an out-of-fold estimate $\widehat{N}_X$; with residual $e = N - \widehat{N}_X$, each of 10,000 null draws forms $N^{(b)} = \widehat{N}_X + \mathcal{P}_b e$, where $\mathcal{P}_b$ permutes the residuals among galaxies. The prediction model is then rerun with $N^{(b)}$. This preserves the estimated dependence of N on the declared baseline while breaking its residual alignment with T . The original unconditional permutation result is retained as a reference, but the residual-permutation result is the stricter incremental-information sensitivity reported below. Repeating the ten-fold comparison over 100 deterministic fold seeds provides a second sensitivity to the particular fold assignment.

5.4 Post-validation universal- $a_\star$ ablation

After the primary Structural Mode result had been established, a targeted ablation was added to test a stronger interpretation: whether the structural predictive power of N depends critically on allowing each galaxy to carry its own inferred $a_\star$. This ablation was not part of the initial fixed validation program and is reported as a post-validation falsification test.

Four branches are compared. The authority branch retains the individually inferred $a_\star$ for every galaxy. Three counterfactual branches force one common acceleration scale across the full primary sample:

$$a_{\star, \text{MONDlike}} = 1.2 \times 10^{-10} \text{ m s}^{-2}, \quad (22)$$

$$a_{\star, \text{median}} = 1.09284448 \times 10^{-10} \text{ m s}^{-2}, \quad (23)$$

$$a_{\star, \text{global}} = 1.25268256 \times 10^{-10} \text{ m s}^{-2}. \quad (24)$$

The first equals the standard MOND comparison scale used by the CPTG workbench [4]; here it serves only as a numerical universal-scale control inside the CPTG equation. The second is the median of the 153 individually inferred CPTG values. The third is selected by minimizing total CPTG rotation-curve χ^2 over all 3168 primary-sample points under the restriction that one common $a_\star$ must be used by every galaxy. Morphology and other validation metadata do not enter that optimization.

For each fixed-scale branch, R_c , the nonlinear field, C_g , λ , N , and the CSMI label are recomputed from the raw SPARC source rows. This gives the universal-scale restriction its strongest kinematic opportunity while separately measuring what happens to the downstream Structural Mode.

The term “fixed” refers to the theory and implementation used for this campaign; it is not a claim that the entire statistical analysis was prospectively preregistered before any SPARC exploration. The genuinely prospective test is future application of the unchanged N definition and CSMI map to an independent survey-quality galaxy sample.

6 Primary Results

6.1 Threshold-independent Structural Mode ordering

The principal threshold-independent test compares empirical Structural Mode with the non-input SPARC Hubble-type coordinate. Across the 153 primary galaxies,

$$\rho_S(N, T) = -0.60817, \quad p = 7.58 \times 10^{-17}, \quad (25)$$

with a galaxy-bootstrap 95% interval approximately

$$-0.7084 \leq \rho_S(N, T) \leq -0.4833. \quad (26)$$

SPARC T increases toward later and irregular systems, so the negative correlation means that larger N corresponds statistically to earlier structural populations.

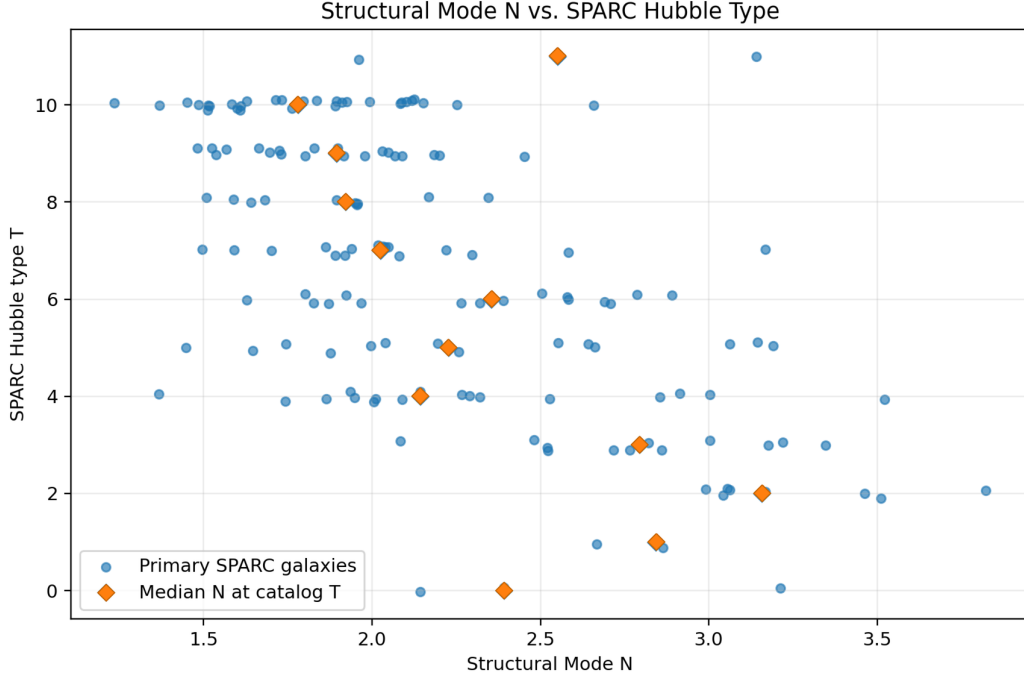


Figure 2: Empirical Structural Mode N against the non-input SPARC Hubble-type coordinate for the 153-galaxy primary sample. The plotted T positions include a small deterministic vertical jitter only to reveal overlapping galaxies; all statistics use the original integer catalog values. Orange diamond markers show the median N at each populated T . Catalog morphology is not supplied to the Structural Mode calculation.

6.2 Non-input SPARC structural observables

Table 3 reports the complete family of catalog validation coordinates tested in the fixed metadata-correlation analysis, including the Benjamini–Hochberg adjusted probability q_{BH} . All ten tested coordinates survive the false-discovery correction. These quantities belong to the same SPARC observational ecosystem and are therefore validation coordinates rather than an external replication sample.

Validation quantity	n	$\rho_S(N, X)$	p	q_{BH}
Effective surface brightness SB_{eff}	153	+0.65869	2.16×10^{-20}	2.16×10^{-19}
Hubble type T	153	-0.60817	7.58×10^{-17}	3.79×10^{-16}
HI radius R_{HI}	149	+0.57442	1.87×10^{-14}	4.96×10^{-14}
3.6 μm luminosity $L_{3.6}$	153	+0.56768	1.98×10^{-14}	4.96×10^{-14}
HI mass M_{HI}	153	+0.55268	1.29×10^{-13}	2.59×10^{-13}
Flat velocity V_{flat}	123	+0.52415	4.91×10^{-10}	8.19×10^{-10}
Disk scale length R_{disk}	153	+0.41087	1.33×10^{-7}	1.89×10^{-7}
Disk surface brightness SB_{disk}	153	+0.38143	1.15×10^{-6}	1.43×10^{-6}
Distance D	153	+0.27043	7.23×10^{-4}	8.03×10^{-4}
Effective radius R_{eff}	153	+0.23896	2.93×10^{-3}	2.93×10^{-3}

Table 3: Complete fixed family of monotonic relations between empirical N and non-input SPARC catalog quantities. Distance is retained because it also serves as an observational sampling/selection control below.

These correlations should not be interpreted as statistical independence from conventional galaxy structure. The opposite is being tested: N is intended to measure physical organization, so it should covary with separately cataloged manifestations of that organization. When $L_{3.6}$, R_{eff} , M_{HI} , R_{disk} , and SB_{eff} are controlled jointly, the residual N - T relation falls to $\rho = -0.14887$ with $p = 0.0663$. On the 123-galaxy subset with measured V_{flat} , controlling jointly for V_{flat} , SB_{eff} , $L_{3.6}$, R_{disk} , and M_{HI} gives $\rho = -0.10584$ with $p = 0.244$. Thus the result is not that N is independent of the galaxy structure it measures; rather, a post-solution CPTG coordinate recovers organization that is also visible in conventional structural measurements.

6.3 Radial-sampling and distance controls

The sampled estimator in Eq. (19) is not invariant to radial sampling density, so the primary structural relation must be checked against the number of measured rotation-curve points. In the primary sample,

$$\rho_S(N, n_{\text{pt}}) = +0.69216, \quad p = 3.84 \times 10^{-23}, \quad (27)$$

$$\rho_S(T, n_{\text{pt}}) = -0.52191, \quad p = 4.57 \times 10^{-12}, \quad (28)$$

$$\rho_S(N, D) = +0.27043, \quad p = 7.23 \times 10^{-4}, \quad (29)$$

$$\rho_S(T, D) = -0.52814, \quad p = 2.29 \times 10^{-12}. \quad (30)$$

Sampling density is therefore associated with both the Structural Mode readout and morphology and cannot be ignored.

Using the same rank-residual partial-Spearman construction as the other control tests

gives

$$\rho_S(N, T \mid n_{\text{pt}}) = -0.40108, \quad p = 2.78 \times 10^{-7}, \quad (31)$$

and controlling jointly for point count and distance gives

$$\rho_S(N, T \mid n_{\text{pt}}, D) = -0.30081, \quad p = 1.58 \times 10^{-4}. \quad (32)$$

The raw relation is reduced by these observational controls but remains statistically significant.

A complementary restriction to increasingly well-resolved rotation curves leaves the raw structural ordering close to its full-sample value:

Minimum radial points	Galaxies	$\rho_S(N, T)$	p
10	115	-0.60278	1.02×10^{-12}
15	81	-0.60659	1.95×10^{-9}
20	58	-0.61755	2.42×10^{-7}

Table 4: Resolution-restricted Structural Mode-morphology test. The correlation remains near -0.61 as increasingly sparse rotation curves are removed.

These controls show that radial sampling contributes to the empirical N distribution, as expected from the registered estimator, but does not account for the full cross-galaxy structural ordering.

6.4 Incremental fold-held-out prediction and sampling sensitivity

The fold-held-out prediction test is retained as a secondary sensitivity rather than the primary validation. Table 5 shows how the incremental contribution of N changes as observational sampling descriptors are added to the baseline. “Metadata-5” denotes $L_{3.6}$, R_{eff} , SB_{eff} , R_{disk} , and M_{HI} ; “Metadata-6” additionally includes measured V_{flat} . Point count and distance are added as $\log_{10} n_{\text{pt}}$ and $\log_{10} D$ under the same fold-wise standardization.

Baseline	n	MAE baseline	MAE + N	ΔMAE	p_{resperm}
Metadata-5	153	1.02419	0.97506	+4.80%	6.00×10^{-4}
Metadata-5 + n_{pt}	153	0.97900	0.96483	+1.45%	2.91×10^{-2}
Metadata-5 + n_{pt} + D	153	0.96230	0.96713	-0.50%	0.514
Metadata-6	123	1.04207	1.00851	+3.22%	5.40×10^{-3}
Metadata-6 + n_{pt}	123	1.04130	1.01519	+2.51%	2.13×10^{-2}
Metadata-6 + n_{pt} + D	123	1.02487	1.01650	+0.82%	8.82×10^{-2}

Table 5: Fixed ten-fold prediction sensitivity. Positive ΔMAE denotes lower Hubble-type MAE after adding empirical N . p_{resperm} is the 10,000-draw residual-permutation sensitivity defined in the statistical protocol.

The original unconditional permutation probabilities for the Metadata-5 and Metadata-6 baselines are 4.00×10^{-4} and 3.60×10^{-3} , respectively; the conditional residual-permutation values in Table 5 remain similar for those declared structural baselines. The increment becomes substantially smaller after point count is included and is not detected once both point count and distance are included. Across 100 deterministic ten-fold partitions, the mean improvement from N is 4.43%, 1.25%, and -0.24% for the three Metadata-5 rows, and 2.62%, 1.19%, and 0.06% for the corresponding Metadata-6 rows. The paper therefore does not treat the original 4.80% fold-held-out gain as independent of observational sampling/selection information.

6.5 CSMI population ordering

The descriptive CSMI bands show a strong population sequence. Because the numerical CSMI boundaries are inherited rather than derived prospectively in this manuscript, this subsection is a descriptive concordance analysis of the pre-existing map. For the broad morphology comparisons below, the SPARC type coordinate is grouped as $T = 0-4$ (early/intermediate spiral), $T = 5-8$ (late spiral), and $T = 9-11$ (irregular/BCD). The threshold-independent $N-T$ statistic in Eq. (25) remains the cleaner structural validation. The full CSMI-band by broad-morphology-family contingency table gives a bias-corrected Cramér association

$$V = 0.418, \quad (33)$$

with ordinary contingency-test probability $p = 6.15 \times 10^{-8}$ and a 10,000-permutation probability $p_{\text{perm}} = 1.00 \times 10^{-4}$ at the available permutation resolution. Because several descriptive bands are sparsely populated, the permutation result is the more relevant significance check for this concordance analysis.

A descriptive broad low/high-mode contrast is particularly transparent. For $N \leq 2.15$, there are 89 primary galaxies with median catalog type $T = 9$; 46 are irregular/BCD systems and 11 are early/intermediate spirals. For $N > 2.80$, there are 28 galaxies with median $T = 3$; only one is irregular/BCD while 22 are early/intermediate spirals. Fisher exact tests give

$$p_{\text{irregular}} = 1.39 \times 10^{-6}, \quad p_{\text{early}} = 1.03 \times 10^{-10}. \quad (34)$$

These probabilities describe concordance of the inherited partition and are not assigned the same inferential status as the continuous-valued $N-T$ result.

Quartile	Median N	Median T	Median SB_{eff}	Median V_{flat}	Median s_{outer}	Median $a_{\star}$
Q1	1.6088	9	29.71	83.6	+0.1619	9.42×10^{-11}
Q2	1.9372	7	69.46	91.4	+0.0942	1.18×10^{-10}
Q3	2.2548	6	178.05	153.2	+0.0341	1.02×10^{-10}
Q4	3.0031	3	900.16	184.6	-0.0338	1.23×10^{-10}

Table 6: Population medians across quartiles of empirical Structural Mode. Surface brightness is in $L_{\odot} \text{ pc}^{-2}$, V_{flat} in km s^{-1} , and $a_{\star}$ in m s^{-2} .

The quartile progression is monotonic in the population medians: the lowest- N quartile has median $T = 9$, while the highest- N quartile has median $T = 3$. Effective surface brightness and V_{flat} increase across the same sequence, while the observed outer rotation-curve slope changes from rising to approximately flat or mildly declining.

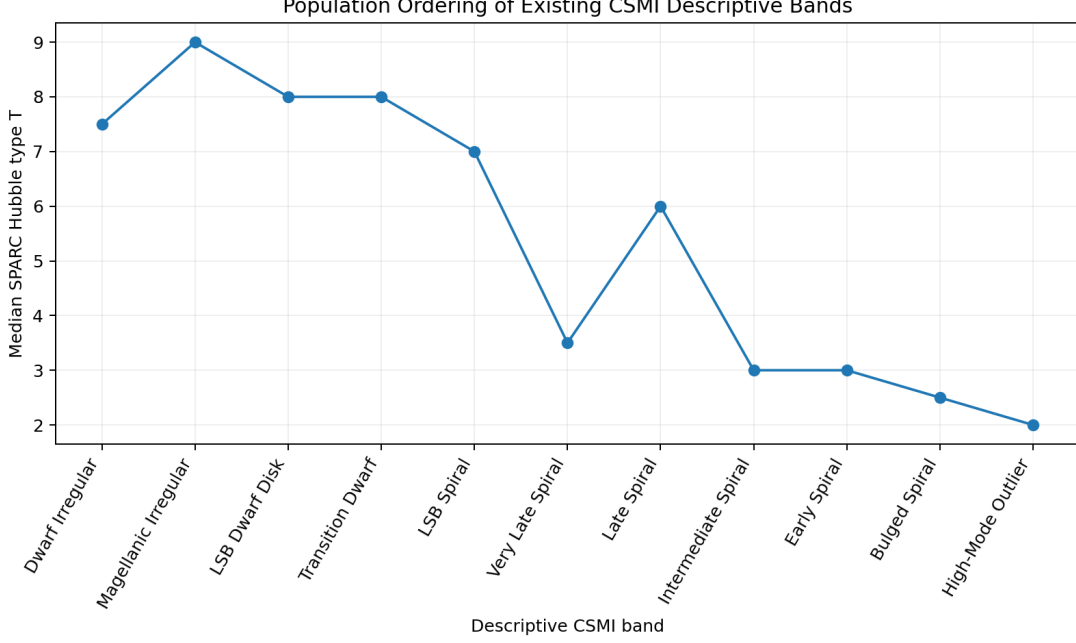


Figure 3: Median SPARC Hubble type by descriptive CSMI interval. Narrowly populated edge bands should be interpreted through the empirical N value and sample count rather than as rigid morphology bins.

6.6 NGC 3741: a catalog-discordance case

NGC 3741 provides an instructive individual example of the catalog-generating role of Structural Mode. The SPARC master catalog retains the conventional type coordinate $T = 10$ and morphology **Im**, placing the object in the irregular/BCD family used in the present validation. The CPTG solution, however, gives

$$N_{\text{NGC 3741}} = 2.65754, \quad (35)$$

which lies in the inherited CSMI interval

$$2.55 < N \leq 2.80 \quad \implies \quad \text{Late Spiral CSMI band.} \quad (36)$$

No catalog morphology is supplied to this calculation. The CSMI label is therefore a statement about the organization of the solved CPTG field, not an imported Hubble classification and not a claim that the conventional catalog must literally be relabeled “Late Spiral.”

Classification channel	NGC 3741 result
SPARC catalog	$T = 10$, Im ; irregular/BCD family
CPTG structural readout	$N = 2.65754$; Late Spiral CSMI interval
Resolved structural evidence	Extremely extended H I disc; H I bar and spiral structure; radial non-circular motions; bar-like young stellar structure and elongated stellar features; recent dynamical work finding susceptibility to global $m = 2$ bar modes [5, 6, 7, 8, 9].

Table 7: NGC 3741 as a catalog-discordance case. The legacy morphology and the CSMI structural label come from different inference channels: the former is externally cataloged, whereas the latter is calculated from the solved CPTG field.

The historical literature makes the disagreement scientifically useful rather than merely anomalous. Early work described NGC 3741 as a nearby gas-rich dwarf irregular with an extraordinarily extended H I disc and, within conventional halo decompositions, as one of the most dynamically dark irregular galaxies known [5]. Subsequent H I modelling found non-circular radial motions consistent with an inner bar and outer accretion [6]. A later direct pattern-speed analysis measured the H I bar and emphasized that the gaseous disc contains both a bar and spiral structure even while the galaxy continued to be described as a dwarf irregular [7].

Deeper optical and H I work sharpened the same picture. The 2022 Smallest Scale of Hierarchy study used Large Binocular Telescope imaging together with archival Hubble Space Telescope data to identify a bar-like stellar structure joined to two elongated young stellar features, while its H I analysis confirmed central spiral arms, radial motions, high central gas density, and a steeply rising inner rotation curve [8]. That study explicitly considered whether NGC 3741 could instead represent an exceptionally low-luminosity spiral system. More recent dynamical work finds the stellar and gaseous discs moderately unstable to global $m = 2$ modes and produces gaseous and weak stellar bars in simulations constrained by the observed system [9].

This chronology is not presented as a temporal prediction by CPTG: the cited observations and analyses precede the present Structural Mode study. Its significance is instead methodological. The morphology-blind CPTG solution assigns NGC 3741 a substantially more organized CSMI band than its legacy **Im** label, and independent resolved observations reveal bar, spiral, and coherent disc structure that make that higher structural organization physically intelligible. NGC 3741 therefore illustrates why solution-derived classification matters: what a galaxy looks like in a legacy morphology catalog and how its solved internal structure is organized need not be the same statement. Structural Mode measures the latter. In a future prospective survey, catalog-discordant systems of this kind would become natural targets for deeper structural follow-up.

6.7 Outer-profile cross-prediction

The observed outer logarithmic velocity slope supplies a second structural diagnostic,

$$s_{\text{outer}} = \frac{d \ln V_{\text{obs}}}{d \ln r} \quad (37)$$

estimated by an ordinary least-squares fit of $\ln V_{\text{obs}}$ against $\ln r$ over the outermost one-third of the available radial points, with at least three valid positive points required. The result is

$$\rho_S(N, s_{\text{outer}}) = -0.44847, \quad p = 6.12 \times 10^{-9}. \quad (38)$$

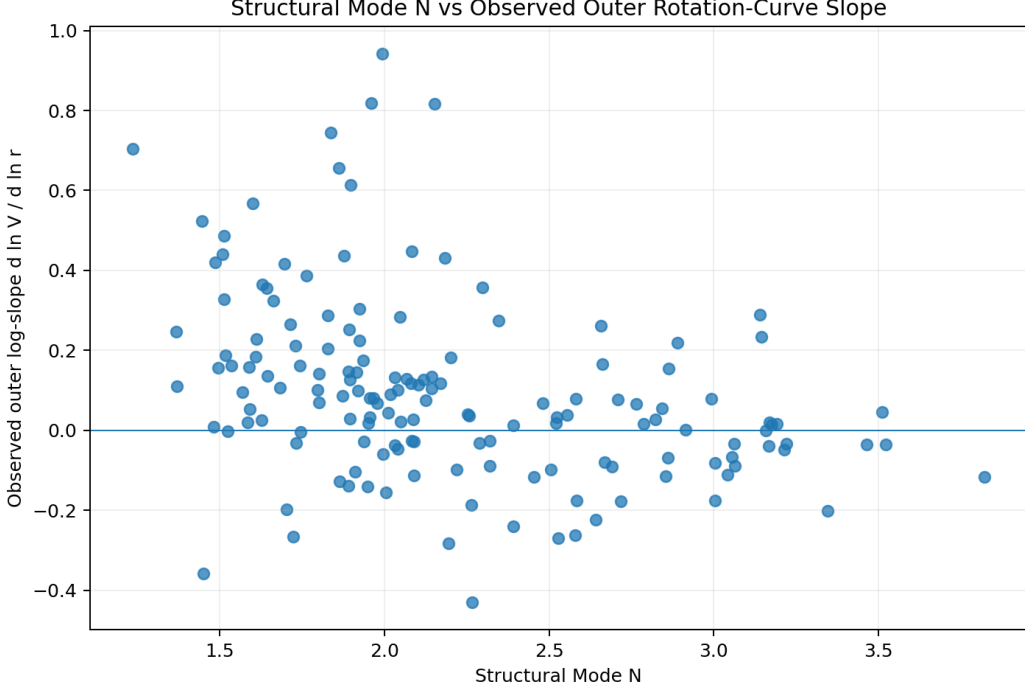


Figure 4: Empirical Structural Mode N versus the observed outer rotation-curve slope. This is a downstream cross-prediction from the same rotation-curve dataset, not an independent external dataset.

Because s_{outer} is calculated from the same rotation-curve measurements used to infer the galaxy field, Eq. (38) is not claimed as statistically independent validation. Its value is structural: the mode orders a coherent change in radial profile behavior.

7 Stability and Classification Boundary

7.1 Measurement-error propagation

For every primary galaxy, a deterministic ten-realization perturbation stress test shifts each observed velocity point independently by its quoted SPARC uncertainty, re-fits a_* , re-solves the CPTG field, and recomputes N . This finite stress test is intended to test local stability to quoted velocity errors, not to serve as a converged posterior uncertainty analysis. Across the 153 galaxies,

$$\text{median } \sigma_N = 5.54 \times 10^{-4}, \quad (39)$$

while the median absolute shift in N is

$$\text{median } |\Delta N| = 3.98 \times 10^{-4}. \quad (40)$$

The mean CSMI-label retention fraction is 98.69%, and 95.42% of galaxies retain the same label in all ten realizations. At the same time, the median relative standard deviation of the re-fitted a_* values is approximately 4.86%. Empirical N is therefore highly stable against the quoted pointwise velocity-error perturbations in this test even when the fitted acceleration scale is allowed to move in every realization.

7.2 Radial sampling

Radial resampling probes a different sensitivity because N is explicitly a structural readout of the resolved radial field. Under leave-one-interior-point-out resampling, 147 galaxies have sufficient radial coverage and the Spearman correlation between the baseline N values and each galaxy's median perturbed value is

$$\rho_S = 0.98397, \quad (41)$$

with median $|N_{\text{pert,med}} - N| = 0.00941$ and mean discrete-label retention 73.05%. Random retention of approximately 80% of radial points while holding the endpoints gives, for 115 eligible galaxies,

$$\rho_S = 0.99076, \quad (42)$$

with median $|N_{\text{pert,med}} - N| = 0.03480$ and mean label retention 50.40%. Thus the population ordering of empirical N is robust even when hard CSMI interval assignments become sensitive to sparse sampling or proximity to a threshold. This distinction is expected for any thresholding of a continuous coordinate and is why the continuous-valued empirical mode is the authoritative scientific readout.

The appropriate reporting hierarchy is consequently

$$\begin{array}{c} \text{empirical } N_{\text{num}} \\ \downarrow \\ \text{CSMI label} \\ \downarrow \\ \text{threshold margin or confidence indicator} \end{array} \quad (43)$$

This recommendation does not alter the theoretical definition of N or the current interval map; it records the empirical uncertainty of downstream classification near an interval boundary.

8 Acceleration-Scale Separation and Ablation

The Structural Mode result is not organized around the acceleration scale, but the two quantities should be distinguished explicitly. Across the 153 primary galaxies,

$$\rho_S(N, a_\star) = 0.09987, \quad p = 0.219. \quad (44)$$

The fitted scale itself shows little monotonic ordering with the principal structural coordinates,

$$\rho_S(a_\star, T) = -0.06308, \quad p = 0.439, \quad \rho_S(a_\star, s_{\text{outer}}) = -0.08059, \quad p = 0.322. \quad (45)$$

Controlling for $a_\star$ leaves the Structural Mode–morphology relation essentially unchanged,

$$\rho_S(N, T \mid a_\star) = -0.60610, \quad p = 1.03 \times 10^{-16}. \quad (46)$$

Thus empirical N is not a monotonic repackaging of the fitted CPTG acceleration-response scale.

8.1 Universal- $a_\star$ ablation

The universal-scale ablation separates two questions: whether one common acceleration scale can reproduce the galaxy kinematics, and whether the downstream Structural Mode depends strongly on the numerical value of that scale. Table 8 gives the result.

$a_\star$ branch	χ^2/N_{pt}	$\rho_S(N, T)$	$\rho_S(N, s_{\text{outer}})$	Cramér V
Object-specific	8.6419	−0.6082	−0.4485	0.4179
MOND-like fixed	47.7547	−0.5972	−0.4380	0.3982
Sample-median fixed	49.9368	−0.5986	−0.4392	0.4002
Global- χ^2 fixed	47.5072	−0.5966	−0.4372	0.3955

Table 8: Universal- $a_\star$ ablation. The globally optimized fixed scale strongly worsens the rotation-curve likelihood but leaves most of the Structural Mode population ordering intact.

The best universal scale is

$$a_{\star, \text{global}} = 1.25268256 \times 10^{-10} \text{ m s}^{-2}, \quad (47)$$

and gives

$$\chi_{\text{fixed}}^2 = 150502.72, \quad \chi_{\text{object}}^2 = 27377.46, \quad (48)$$

so that

$$\frac{\chi_{\text{fixed}}^2}{\chi_{\text{object}}^2} = 5.4973. \quad (49)$$

The object-specific branch contains 153 galaxy-level acceleration-scale parameters whereas the universal branch contains one. For the common adopted point-error likelihood, the information criteria are evaluated as

$$\text{AIC} = \chi^2 + 2k, \quad \text{BIC} = \chi^2 + k \ln n, \quad n = 3168, \quad (50)$$

with $k = 153$ for the object-specific $a_\star$ branch and $k = 1$ for the universal restriction; all other registered galaxy-law quantities are fixed in both branches. The preference remains overwhelming after that parameter-count penalty:

$$\Delta \text{AIC}_{\text{fixed-object}} = 1.22821 \times 10^5, \quad (51)$$

$$\Delta \text{BIC}_{\text{fixed-object}} = 1.21900 \times 10^5. \quad (52)$$

These criteria establish a relative preference between the two CPTG acceleration-scale architectures under the same adopted point-error likelihood; they are not an assertion that either branch supplies an absolute goodness-of-fit calibration. Within that comparison, one universal acceleration scale is strongly disfavored by the primary-sample kinematics.

The Structural Mode itself behaves very differently. Comparing the globally optimized fixed branch with the object-specific authority branch gives

$$\rho_S(N_{\text{object}}, N_{\text{fixed}}) = 0.99808, \quad (53)$$

with median $|\Delta N| = 0.00510$ and 93.46% retention of the discrete CSMI label. The fixed branch retains 98.1% of the magnitude of the N – T rank correlation, 97.5% of the N –outer-slope correlation, and 94.7% of the bias-corrected Cramér association between CSMI band and broad morphology family. Adding fixed-branch N to the five-variable metadata predictor continues to improve fold-held-out Hubble-type prediction by about 5.10% rather than collapsing.

Figure 5 shows the point-by-point Structural Mode comparison between the object-specific and globally optimized universal-scale branches.

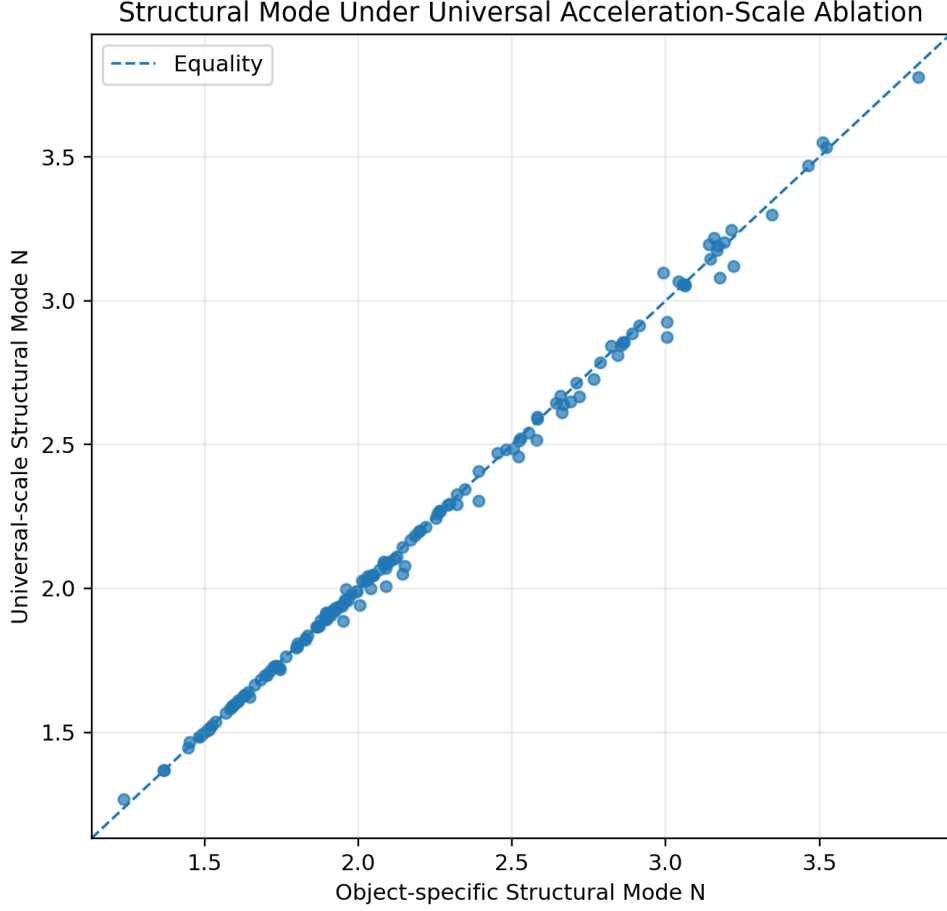


Figure 5: Object-specific Structural Mode N versus the mode recomputed after imposing the globally optimized universal a_* . The diagonal denotes equality. The near-diagonal population, $\rho_S = 0.99808$, makes the structural invariance of the ablation directly visible even though the universal-scale branch has a much poorer kinematic likelihood.

This result falsifies the stronger hypothesis that Structural Mode validates the object-specific nature of a_* because N would fail if one universal scale were imposed. The actual result is more strongly decoupled than that hypothesis assumed. Galaxy-specific a_* values are required by the kinematic likelihood, while the ordering measured by N is remarkably insensitive to replacing those values by a common scale. Structural Mode therefore validates radial organization encoded by the solved CPTG field, but it is not by itself an independent test that a_* must vary from galaxy to galaxy.

The near-invariance is consistent with the normalization insensitivity built into Structural Mode. Equation (11), with $s = 1$, removes any purely affine normalization of C_g from N . Replacing the object-specific acceleration scale by a common value is not an affine transformation of the solved field, so the ablation does not uniquely decompose the resulting field change. What it establishes is that the remaining profile changes are insufficient to substantially reorder the structural centroid measured by N . Kinematic amplitude and structural organization can therefore separate sharply: the former is badly degraded while the latter remains ordered.

9 Interpretation

The central result is a change in classification authority. CPTG does not reproduce the SPARC morphology sequence because those labels were supplied to the field solution; they were not. The solved galaxy generates N , N generates a CSMI class, and the external morphology catalog is consulted only afterward. The observed agreement therefore tests whether a theory-generated structural catalog can recover organization that conventional astronomy describes from appearance. In this sense, morphology changes from an input description to an external comparison coordinate.

The first result is structural. Empirical N provides a stable ordering of SPARC galaxies that is strongly related to non-input morphology and other catalog measures of organization. The sampled estimator is demonstrably affected by radial sampling density, but explicit point-count and distance controls leave a significant N – T relation, and increasingly well-resolved subsets retain a raw correlation near -0.61 . The observed structural sequence therefore cannot be attributed solely to the number of measured rotation-curve points.

Second, direct photometric and baryonic variables share much of the same information. Once several such quantities are controlled simultaneously, much of the raw N – T relation is absorbed. This is physically consistent with the interpretation of N as an organization coordinate rather than a statistically independent observable. The fold-held-out prediction test carries the same boundary: N adds measurable information beyond the declared structural-metadata baseline, but the increment shrinks when sampling density is included and is not detected in the full-sample model once both point count and distance are supplied. The direct rank-ordering and resolution-restricted tests are therefore the stronger evidence in the present dataset.

Third, the descriptive CSMI labels remain operational rather than foundational. They map intervals of the continuous-valued empirical mode into a familiar structural vocabulary, but the present catalog is not asserted to be a permanently closed taxonomy. Larger and more diverse samples may resolve additional mode populations and motivate subdivision or refinement of the named CSMI classes, with any such changes derived from the solved N distribution rather than imported from conventional morphology. The numerical thresholds used in the present validation are inherited from the existing workbench and their morphology concordance is not treated as an independent derivation. Measurement-error perturbations preserve the mode very strongly; radial resampling preserves the cross-galaxy ranking more strongly than the hard labels, supporting the reporting order empirical N first, CSMI label second, and threshold/sampling robustness for boundary cases.

Fourth, a_* and N constrain different aspects of the CPTG galaxy solution. The acceleration scale is constrained primarily by kinematic likelihood, whereas Structural Mode reads radial organization. The universal-scale ablation strongly degrades the rotation-curve likelihood while leaving the N ordering nearly unchanged. That result is useful because it prevents the Structural Mode paper from conflating amplitude calibration with structural organization.

10 Future Structural Resolution Beyond Structural Mode

Structural Mode N is intentionally a compact global coordinate. Its success in separating low-mode from high-mode galaxy populations does not imply that it exhausts the structural information contained in the solved CPTG field. On the contrary, the mathematical definition shows that finer information is necessarily compressed by the present scalar readout.

For the continuous Structural Mode functional of Eq. (8), let the normalized radial structural weight over the registered interval be

$$p_r(r) = \frac{w(r)}{\int_{R/5}^R w(r) dr}, \quad \int_{R/5}^R p_r(r) dr = 1. \quad (54)$$

With the dimensionless radius

$$x = \frac{r}{R}, \quad (55)$$

the corresponding dimensionless probability density is

$$p_x(x) = R p_r(Rx) = \frac{R w(Rx)}{\int_{R/5}^R w(r) dr}, \quad \int_{1/5}^1 p_x(x) dx = 1. \quad (56)$$

The continuous Structural Mode can then be written exactly as

$$N = \frac{1}{\langle x \rangle}, \quad \langle x \rangle = \int_{1/5}^1 x p_x(x) dx = \frac{\lambda}{R}. \quad (57)$$

Thus the continuous galaxy Structural Mode retains the mean radial location of the solved structural-response distribution while discarding details of its shape. Distinct structural distributions can therefore possess the same N whenever they have the same weighted radial centroid. A narrowly concentrated response, a broad response, or a response with several separated concentrations may occupy the same global Structural Mode while differing substantially in internal organization.

This immediately suggests a possible family of higher-order post-solution coordinates. For example, the radial width

$$\sigma_x^2 = \int_{1/5}^1 (x - \langle x \rangle)^2 p_x(x) dx \quad (58)$$

would distinguish compact from radially distributed structural response at fixed N . Higher moments could similarly characterize asymmetry, concentration, or tail structure. Explicit multimodality, however, would require resolving multiple peaks or otherwise retaining more of the structural-weight profile; it is not uniquely determined by a finite set of low-order moments. Such quantities would not replace Structural Mode. They would resolve information that the global coordinate intentionally compresses.

Any empirical higher-order coordinate would also require its own registered sampled-data estimator and sampling-density validation. The continuum moments above cannot be identified automatically with a discrete SPARC implementation, just as the registered N_{num} estimator in Eq. (19) is not assumed to be a quadrature rule for the continuous centroid.

A second and potentially more important extension concerns non-axisymmetric structure. The present galaxy realization of N is radial and therefore cannot uniquely identify intrinsically azimuthal features such as bars, spiral-arm multiplicity, lopsidedness, or pitch structure. These features may influence the radial solved field indirectly, but a scalar radial centroid cannot retain their full angular information.

A future non-axisymmetric CPTG solution could test whether an appropriately derived structural field

$$C_g(r, \phi) \tag{59}$$

admits an angular structural decomposition. Schematically, a corresponding structural response could be expanded as

$$w(r, \phi) = \sum_m w_m(r) e^{im\phi}, \tag{60}$$

where the axisymmetric component would describe global radial organization while nonzero angular modes could encode finer departures from axisymmetry. A detected $m = 2$ component alone would not distinguish a bar from a two-armed spiral; that discrimination would require the radial amplitude and phase structure of the mode, together with a separately derived connection to resolved galaxy structure. Equation (60) is therefore a prospective mathematical direction rather than an established CPTG galaxy law; the appropriate two-dimensional structural invariant, normalization, and observable mapping must be derived before such modes can be assigned physical meaning.

NGC 3741 provides a useful motivation for this extension. Its global Structural Mode places the galaxy substantially farther along the organized CSMI sequence than its legacy irregular classification suggests, while resolved observations reveal bar and spiral structure. The present result does not establish that Structural Mode predicts those individual features. It raises the narrower and testable question of whether additional structural information already present in the solved CPTG field is being compressed by the scalar N readout.

The prospective structural readouts therefore form parallel branches rather than a sequential hierarchy:

$$\begin{array}{ccc}
 & \text{solved CPTG field} & \\
 \swarrow & \downarrow & \searrow \\
 \text{global radial} & \text{higher radial} & \text{non-axisymmetric} \\
 \text{organization } N & \text{structural moments} & \text{structural modes}
 \end{array} \tag{61}$$

The present paper establishes only the global radial branch through N . Whether additional information in the solved field can support higher radial coordinates, resolve bars or spiral

structure, or identify other substructure is left as a separate derivation and observational test.

11 Falsifiability and Prospective Replication

The structural interpretation of N would be directly stressed by any of the following outcomes:

1. a future independent survey-quality galaxy sample shows no reproducible relation between empirical N and non-input structural observables;
2. improved radial data or explicit sampling controls cause the population ranking of N to become unstable or remove the structural relation;
3. sampling-controlled fold-held-out analyses consistently show no incremental information where comparable structural metadata are available;
4. an independent survey-quality galaxy sample no longer shows a decisive kinematic preference for object-specific a_* over the best universal-scale CPTG restriction;
5. the CSMI population ordering disappears when the same pre-existing interval map is applied prospectively;
6. a nominally common Structural Mode definition fails when a separately derived organization length is constructed in another CPTG bound-system regime.

No cross-survey replication is claimed in the present paper. A suitable independent rotation-curve sample is not currently available at SPARC-comparable resolution, so the prospective replication is deferred to BIG-SPARC or another future dataset capable of carrying the unchanged calculation without changing Eq. (8), the galaxy field law, or the CSMI intervals. This creates a clean future test because that dataset has not been used to construct the present results.

The final cross-regime test is separate. The present paper validates the galaxy realization of $N = R/\lambda$. It does not assume that the current cluster response-load coordinate is already the same quantity. A cluster-scale Structural Mode must be rederived from the cluster’s own solved organization scale before a cross-regime identification is made.

12 Conclusion

The principal result of this paper is that CPTG can generate a structural classification from the solved galaxy rather than import one from a morphology catalog. Structural Mode N is calculated without Hubble type or visual morphology as an input, yet in 153 primary-SPARC galaxies the empirical sampled mode strongly orders the non-input catalog morphology, $\rho_S(N, T) = -0.60817$, and tracks multiple non-input descriptions of galaxy organization. Radial sampling density is an important property of the registered estimator, but it does not explain the full result: controlling for point count and distance leaves a significant relation, and samples restricted to at least 10, 15, or 20 measured radial points retain $\rho_S(N, T) \simeq -0.60$ to -0.62 .

The evidence also defines the correct hierarchy of claims. CSMI is a descriptive concordance layer on top of the continuous-valued empirical mode, not a morphology input or an independently derived Hubble classification. Fold-held-out prediction improves when N is added to the declared photometric/baryonic baselines, but that increment is reduced by observational sampling controls and is not the primary validation result. Measurement-error and radial-resampling tests instead show that the cross-galaxy ordering of empirical N is robust even when individual threshold labels become sensitive near CSMI boundaries.

The individual NGC 3741 case shows what that structural ordering can mean operationally: the morphology-blind CPTG solution occupies a substantially more organized CSMI interval than the legacy irregular label, while independent resolved observations reveal bar and spiral organization. This is a non-input concordance example rather than a prospective prediction, but it demonstrates why catalog discordance can itself be scientifically informative.

For structural classification inside CPTG, the hierarchy is therefore explicit: measured galaxy data are inputs, the field solution is the physical calculation, and the structural label is an output. The appearance-based catalog label is not required to determine the result.

The acceleration scale a_* remains a separate CPTG coordinate. Its galaxy-specific values are strongly preferred by the kinematic likelihood over a one-scale restriction, while the Structural Mode ordering survives that ablation almost unchanged. The present evidence therefore supports N as a post-solution readout of galaxy organization whose strongest current test is its sampling-robust structural ordering inside SPARC. Prospective cross-survey replication remains for a future comparably resolved independent sample.

Data, Code, and Evidence Availability

The public CPTG repository contains the SPARC workbench implementation and companion CPTG theory papers at <https://github.com/CLG2025/CPTG>. The cumulative evidence package accompanying this manuscript is

`CPTG_StructuralModeN_Primary153_Validation_20260829_r03.zip`

This package preserves the exact workbench source used here, official SPARC metadata, the 153 primary rotation-curve files, per-galaxy solved outputs, statistical tables, multivariate controls, robustness diagnostics, the universal- a_* ablation, and executable verification scripts required to reproduce the reported Structural Mode results. The sampling-control companion is

`CPTG_StructuralModeN_SamplingControls_20260829_r01.zip`

The sampling-control companion additionally preserves the point-count and distance controls, resolution-restricted tests, fixed-fold prediction sensitivities, residual-permutation results, and the script used to reproduce those extensions. The SPARC observational database retains its original provenance and citation [1].

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